1. Static Image Mode (Precision Analysis)

This version is designed for high-resolution scene analysis where maximum detail is required. Combines SegFormer-b5 and YOLOv8l (Large model) for superior accuracy. The user captures a still photo, and the system performs a deep scan to identify every obstacle and path detail. Provides a comprehensive summary of a complex environment before the user starts walking.

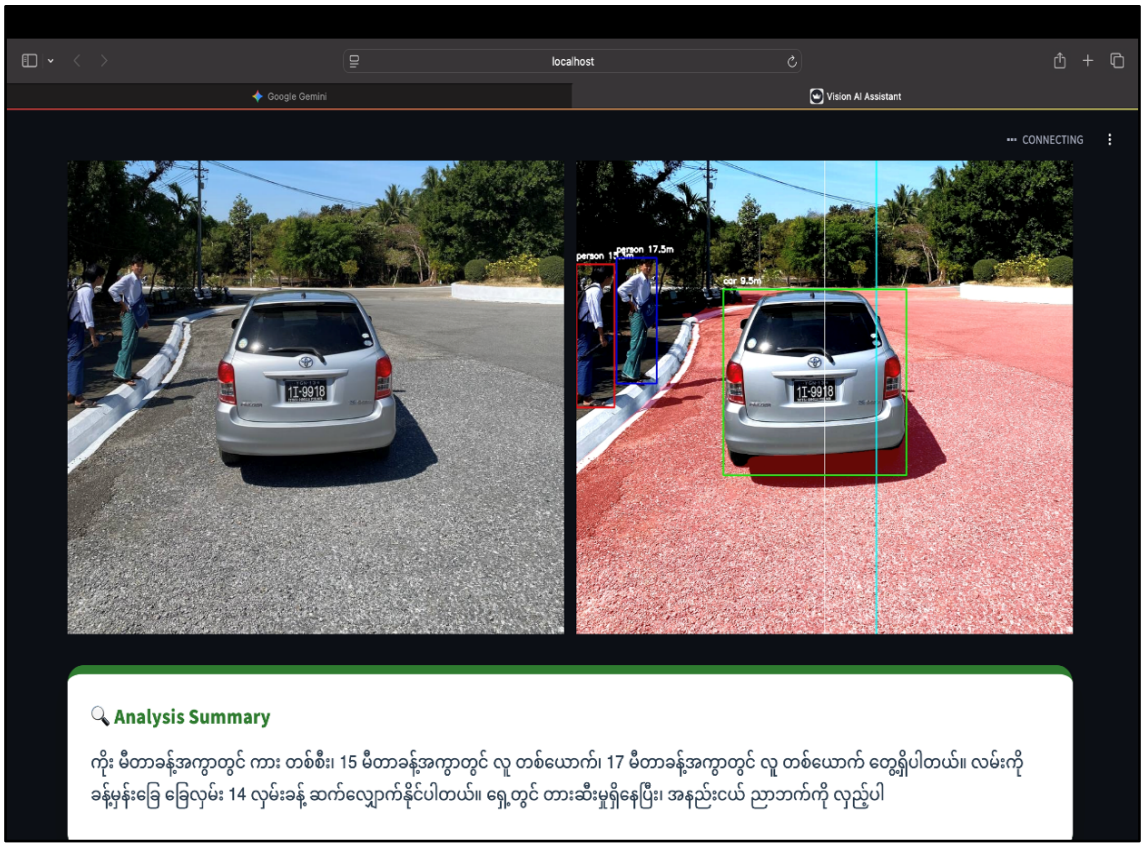


Figure1.1 Safe path detected around vehicle obstacle

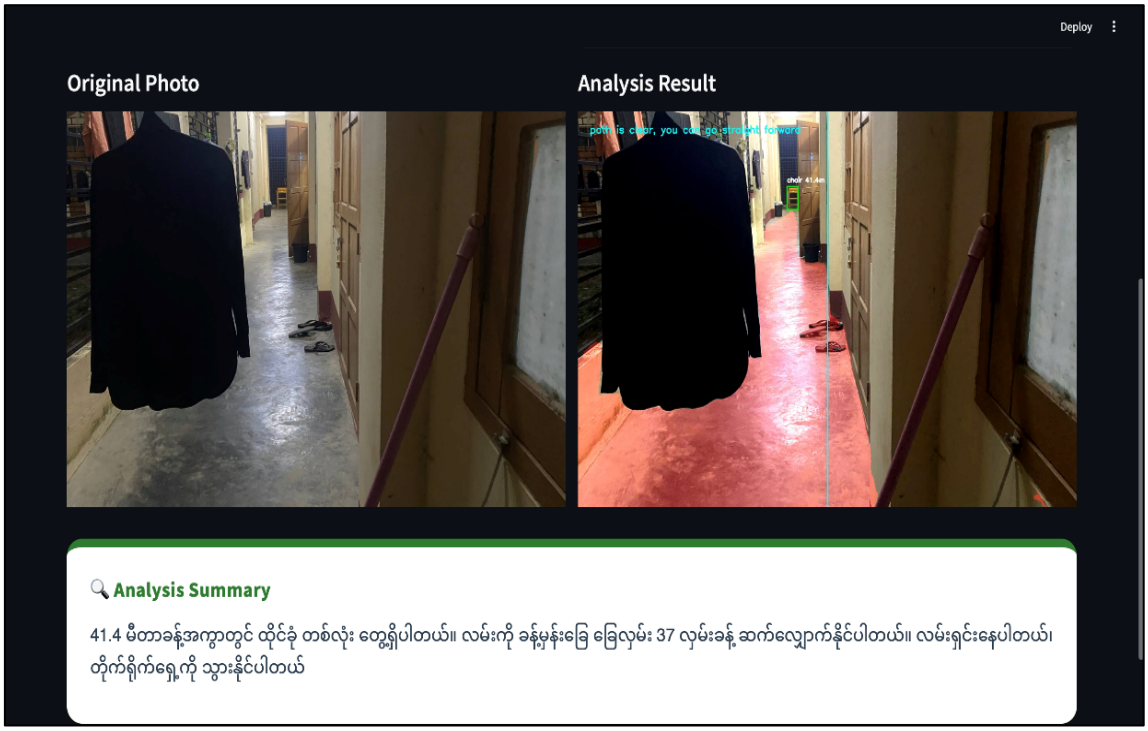


Figure1.2 Narrow Corridor Path Detection & Distant Obstacle Identification



Figure1.3 Road lane segmentation under simulated heavy rain



Figure1.4 Crowded Walkway Navigation & Multi-Person Tracking



Figure1.5 Successful identification of upside-down human subjects and walkable paths

2. Video Analysis Mode (Route Review)

This version allows for the processing of pre-recorded video files to review specific paths.

Powered by a Unified Engine using SegFormer-b3 and YOLOv8m (Medium model). It scans recorded footage and generates navigation summaries as if the users were moving in real-time. Excellent for route familiarity training and system testing to ensure consistent guidance.

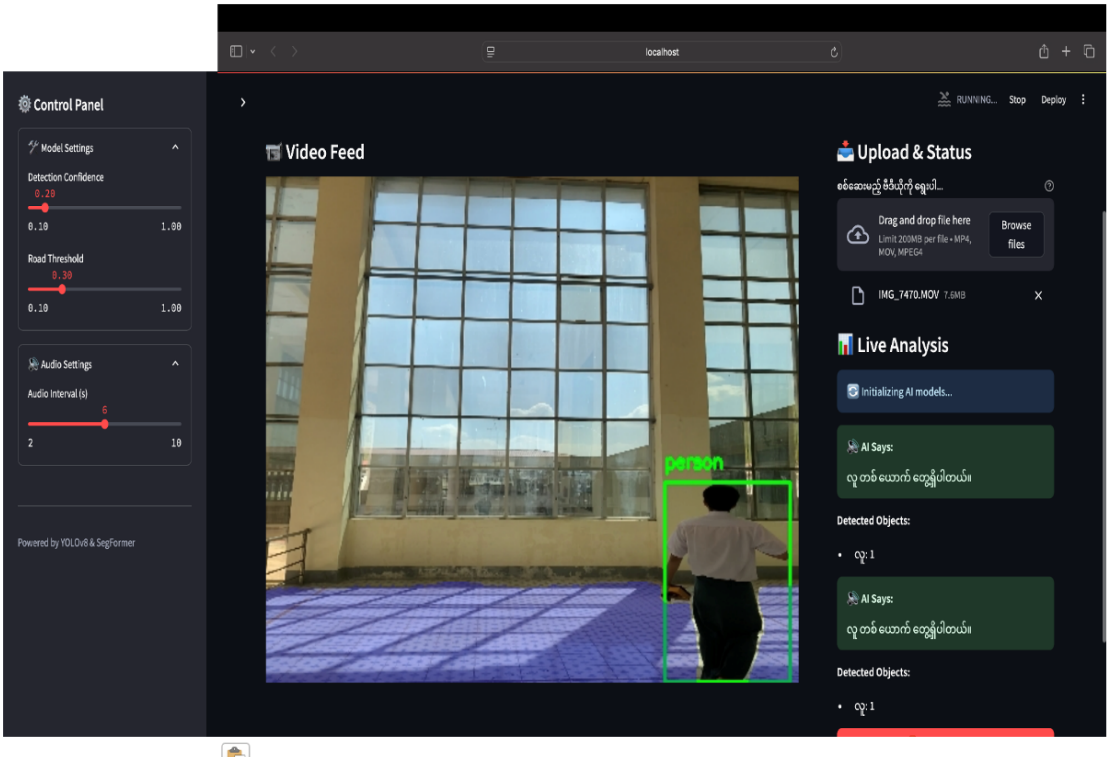


Figure2.1 Indoor Lane Segmentation & Person Tracking

3. Live Vision Mode (Active Real-Time Guidance)

This is the core version for active navigation during daily travel. This uses the same high-speed Unified Engine (SegFormer-b3 + YOLOv8m) for smooth processing. It analyzes a live camera feed continuously, adapting instantly to changing surroundings. Features a Performance Slider (0.1 - 1.00). Users can adjust the processing load (e.g., setting it to 0.2 for a 20% workload) to ensure the AI runs perfectly on any laptop.

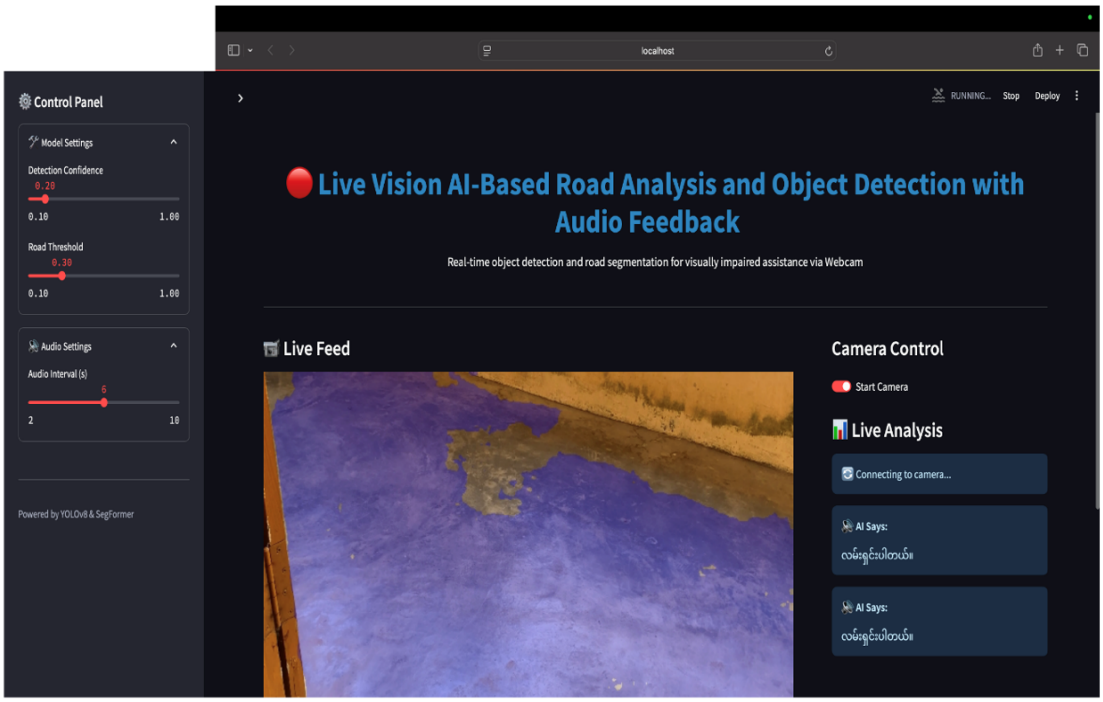


Figure3.1 Live Version Road Detection